SRIKANTH YELEM

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Summary

Electrical Engineer with 4+ years of experience delivering power system, substation, and building electrical solutions across consulting, utility, and EPC environments. Applied load flow, short-circuit, arc flash, and protection coordination studies to improve system reliability, fault selectivity, and safety performance. Led renewable integration, SCADA-based monitoring enhancements, and substation design initiatives, strengthening grid visibility and operational efficiency. Strong expertise in IEC/IEEE standards, grounding design, and equipment engineering, ensuring compliant, resilient, and maintainable electrical infrastructure across project lifecycle.

Skills

Power Systems Engineering: Load Flow Analysis, Short Circuit Studies, Stability Analysis, Power System Modeling, Contingency Analysis, Power Quality Analysis, Renewable Energy Integration (Solar/Wind), Grid Interconnection Studies

Protection, Relaying & Substation Automation: Distance, Differential, Overcurrent & Earth Fault Protection, Protection Coordination Studies, Relay Setting Calculations, IEC 61850 Protocol, SCADA Systems, Fault Analysis & Troubleshooting

Renewable Integration & Grid Modernization: Inverter-Based Resource Modeling, Distributed Energy Resource (DER) Interconnection Studies, Grid-Forming & Grid-Following Control Strategies, Advanced Distribution Management Systems (ADMS)

Engineering Tools & Software: PSCAD (Power System Transients), ETAP, PSS®E, OpenDSS, PowerWorld, ASPEN OneLiner, MATLAB/Simulink, Typhoon HIL Simulator (Real-Time Simulations), AutoCAD Electrical, Advanced Microsoft Excel

Professional Experience

Electrical Engineer-Microgrid Solutions(*Intern*), *Delta Electronics* June 2026 – Present | NC, USA

- Implemented PSCAD to develop a complete control model of a 3.44 MVA / 690 V MV inverter, implementing grid-forming operation with P-f and Q-V droop, virtual synchronous machine (virtual inertia), and island-mode voltage and frequency regulation for microgrid applications.
- Designed and tuned the full control stack — SRF-PLL, dq current loops, and active/reactive power regulation — and implemented virtual-impedance fault-current limiting (1.1–1.5 pu) to ensure stable inverter behavior under weak-grid and fault conditions.
- Engineered a seamless grid-following-to-grid-forming transition using angle pre-synchronization and mode-switching logic, validating uninterrupted islanding through time-domain simulation — a core requirement for microgrid resilience.
- Analyzed a 215 kW grid-connected Battery Energy Storage System power conversion system in PSCAD, including a 1.2 kV split DC link, dual parallel three-level ANPC converters, LCL filtering, measurement circuits, PWM generation, and protection logic.
- Developed a structured 32-case commissioning and validation plan covering DC-link energization, zero-power synchronization, active and reactive power control, converter current sharing, overcurrent protection, charging operation, grid faults, LVRT/HVRT, anti-islanding, and DC-breaker events.
- Evaluated controller and switching interfaces using Fortran-based system and module controller wrappers, 16 kHz PWM logic, duty-cycle limiting, and cycle-by-cycle current protection, while defining expected waveforms, monitored signals, and pass/fail criteria for each test (Under Implementation).

Electrical Engineer, *VRVS India Pvt. Ltd* Jan 2016 – Dec 2017 | AP, India

- Performed load flow, short-circuit, and stability studies using PSS®E and PSCAD for a 132 kV transmission network expansion, improving voltage stability and ensuring reliable grid operation under varying demand conditions.
- Conducted DER interconnection studies using OpenDSS for 60 MW solar integration at distribution level, evaluating voltage fluctuations, fault levels, and ensuring compliance with grid codes for stable network performance.
- Developed inverter-based resource models in MATLAB/Simulink, evaluating grid-forming and grid-following controls to support renewable integration and improve system stability under dynamic conditions.
- Executed contingency analysis and fault simulations using PowerWorld and PSCAD, identifying overload risks and supporting corrective strategies to maintain system reliability during peak demand and unexpected transmission outages.

Electrical Engineer, *Futronics Ltd.* Oct 2011 – Aug 2013 | AP, India

- Designed 132 kV AIS substations including single line diagrams and layouts using AutoCAD Electrical, ensuring compliance with specifications and optimizing space utilization for installation, operation, and long-term maintenance accessibility.
- Performed transformer and switchgear selection for high-capacity loads, applying insulation coordination principles to ensure reliable equipment performance across transmission and industrial electrical infrastructure projects.
- Developed cable sizing and routing plans for HT and LT networks, optimizing voltage drop, thermal performance, and installation feasibility in compliance with engineering standards and project execution requirements.

Education

Ph.D, *University of Missouri* Jan 2020 – Dec 2026 | MO, USA
Electrical and Computer Engineering

Master of Science, *University of Missouri* Jan 2018 – Dec 2019 | MO, USA
Electrical Engineering